Short-horizon crypto signals need a cost-aware test

Five-minute OHLCV pilot | Separate supplementary project through September 30, 2026

Executive summary

At 20 bps, 0 of 18 configurations had positive January-September 2026 net returns. Across all four partitions, 72 of 72 primary case-period results were negative. Every case is retained; this is descriptive, not a corrected discovery test.

Momentum L15m / R60m: gross -26.67%; net at 5 bps -98.96%; at 20 bps retained only 2.86e-08 of initial capital (net return rounds to -100%). The selected 2024 control was already negative after costs.

Reversal L15m / R60m: gross 33.53%; net at 5 bps -98.11%; at 20 bps retained only 5.22e-08 of initial capital (net return rounds to -100%). The selected 2024 control was already negative after costs.

Five-minute sampling is a new measurement granularity, not new untouched market history. Every period in this follow-up, including 2026, is exploratory/retrospective. Shorter horizons do not lower costs, and OHLCV closes do not demonstrate executable microstructure profits.

Pilot dimension	Declared scope
Market	Eight fixed Binance spot USDT pairs; theoretical long/short
Grid	Momentum/reversal x 5/15/30-minute lookback x 5/15/60-minute rebalance = 18
Primary	20 bps full-L1 cost, one five-minute-bar execution lag
Stress	0/5/10/15/40 bps; extra five-minute execution delay at 20 bps
Evidence	All four periods are exploratory/retrospective; no fresh holdout

This is an educational close-price simulation, not risk-free arbitrage or a deployment recommendation. The original project and its expanded hourly extension remain unchanged.

02 / Actual five-minute data and granular halt audit

Partition	Dates	Purpose
Warm-up	December 2021	Signals only
Development	2022-2023	Implementation and historical comparison
Selection	2024	One control per family by net daily Sharpe
Retrospective	2025	Already examined at coarser granularity
Retrospective 2026	Jan 1-Sep 30	273 days; 78,624 regular five-minute intervals

Source and timestamp conventions

The study uses actual Binance 5m archives, not prices interpolated from hourly bars. Each completed bar is indexed by open time plus five minutes; Binance spot timestamp units change from milliseconds to microseconds in January 2025. Archive checksums, metadata anomalies, repairs, duplicates and unexplained gaps are audited before research.

Data-audit status: pass_with_documented_exceptions. This is not an exception-free clean pass. Observed asset-bar rows: 4066432; expected: 4066560; missing asset-bars: 128. The 464 verified monthly archives cover 178.7 MB compressed. Sixteen close-time metadata anomalies comprise one SOL metadata-field-only repair and 15 warnings; no OHLCV price/volume is repaired. Exact counts and digests are retained.

March 24, 2023: one observed 160-minute bridge

The granular suspension treatment is 11:25 to 14:05 UTC, not the hourly study's coarser 11:00 to 15:00 window. Raw partial and zero-trade records are retained as source evidence, but no interior price valuation or execution is simulated. Existing holdings earn the single true close-to-close return over 160 elapsed minutes. Sixteen missing five-minute endpoints per asset (12:45-14:00 UTC) are documented; missing prices are never filled. No intrahalt mark-to-market drawdown can be reconstructed.

Hourly reconciliation is a data check, not a portfolio claim

Among 338,864 finite hourly price comparisons, three close discrepancies and 134 quote-volume discrepancies are retained, all in December 2021 warm-up; none occur among 332,920 finite evaluated-window close comparisons in January 2022-September 2026. Verified five-minute daily closes match the hourly source at the three discrepant endpoints but differ from the monthly five-minute archives. No source values are replaced, and no evaluated return or at-most-30-minute signal uses those early-December endpoints. The data check does not imply equal portfolio returns across trading frequencies. Volume tolerance and every exception remain in the receipts.

03 / Declared grid, historical selection, and accounting

Cross-sectional signals

At scheduled UTC decisions, rank each coin's elapsed-time trailing return, demean the ranks and normalize full L1 exposure to one. Momentum holds high-rank coins long and low-rank coins short; reversal flips the signs. Ties use averaged ranks; constant signals remain cash. Targets are dollar-neutral, but market beta need not be zero. Scheduled 5/15/60-minute rebalances do not trade every five-minute observation unless the cadence is five minutes.

Selection is a control, not independent confirmation

One configuration per family is selected by highest defined 2024 net daily-return Sharpe at 20 bps. Ties prefer lower turnover, then fixed experiment ID. Negative winners remain research controls. The complete 18-case, four-period, seven-scenario table prevents hiding unsuccessful cases. The protocol was declared before strategy calculations; source-data inspection was already underway. This is not a data-naive preregistration and does not erase prior knowledge of the same calendar history.

2024 selected control	Net return	Net Sharpe	Full-L1 turnover
Momentum L15m / R60m	-100.00%	-88.41	11416.89
Reversal L15m / R60m	-100.00%	-94.17	11416.89

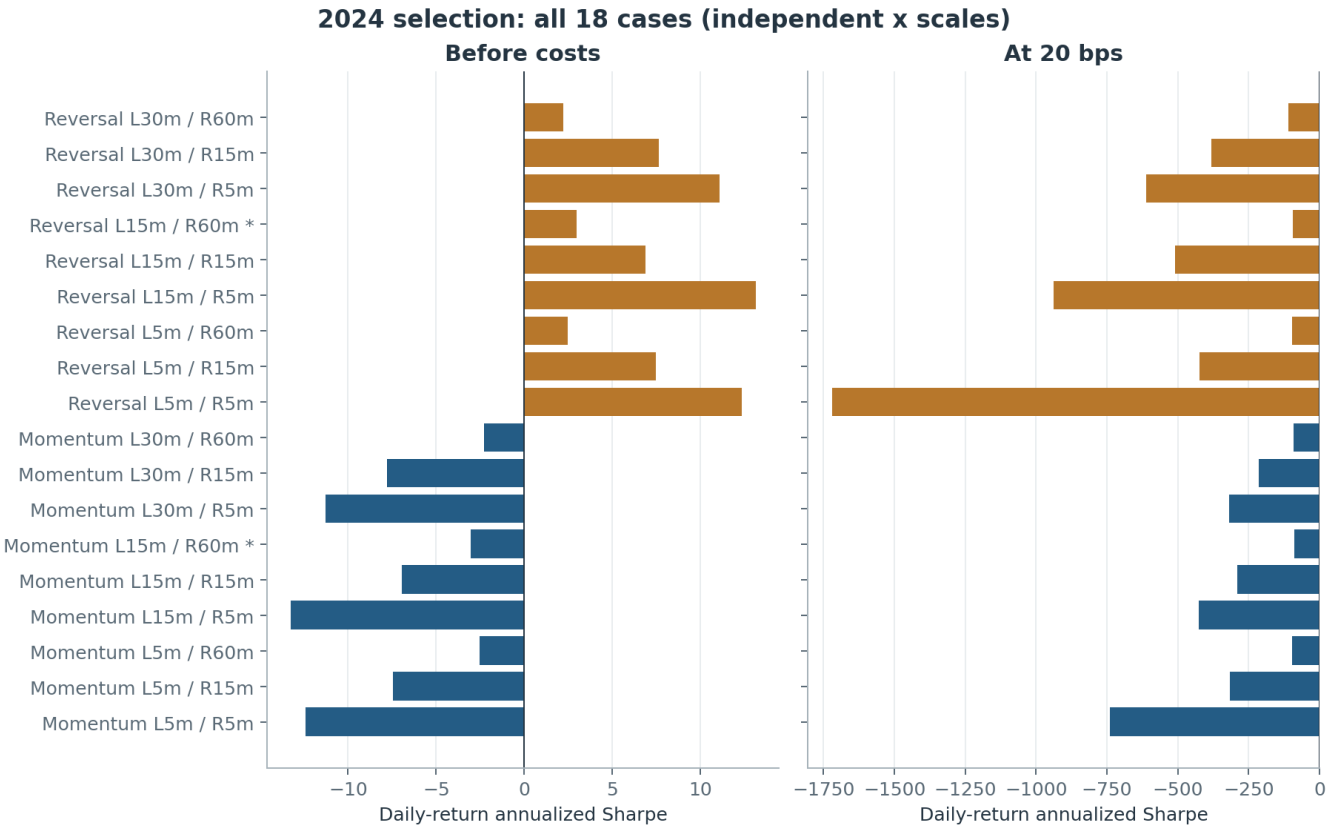
Execution and cash resets

Targets use completed observed endpoints only. Lag one earns the following five-minute interval; lag two adds another five minutes. The suspension prevents interior execution. Each evaluation starts from cash, preserves warm-up signals, and charges both initial entry and final liquidation. No forced cash-out is invented before the 2023 halt.

Net return = held-weight simple asset return - (bps / 10,000) x full-L1 executed weight change. Ordinary position-weight drift is omitted, as in the course model. Equity compounds all interval returns; near-total loss can round to -100% while mathematical equity is still positive. Daily returns compound five-minute and bridge returns using their actual UTC intervals.

Annualization uses 365 calendar days, zero risk-free rate, daily sample volatility and mean-daily-return Sharpe. Nine-month total return differs from CAGR and annualized mean return. BTC alpha/beta use daily OLS with an intercept and HAC(7) uncertainty; these are descriptive, not proof of arbitrage.

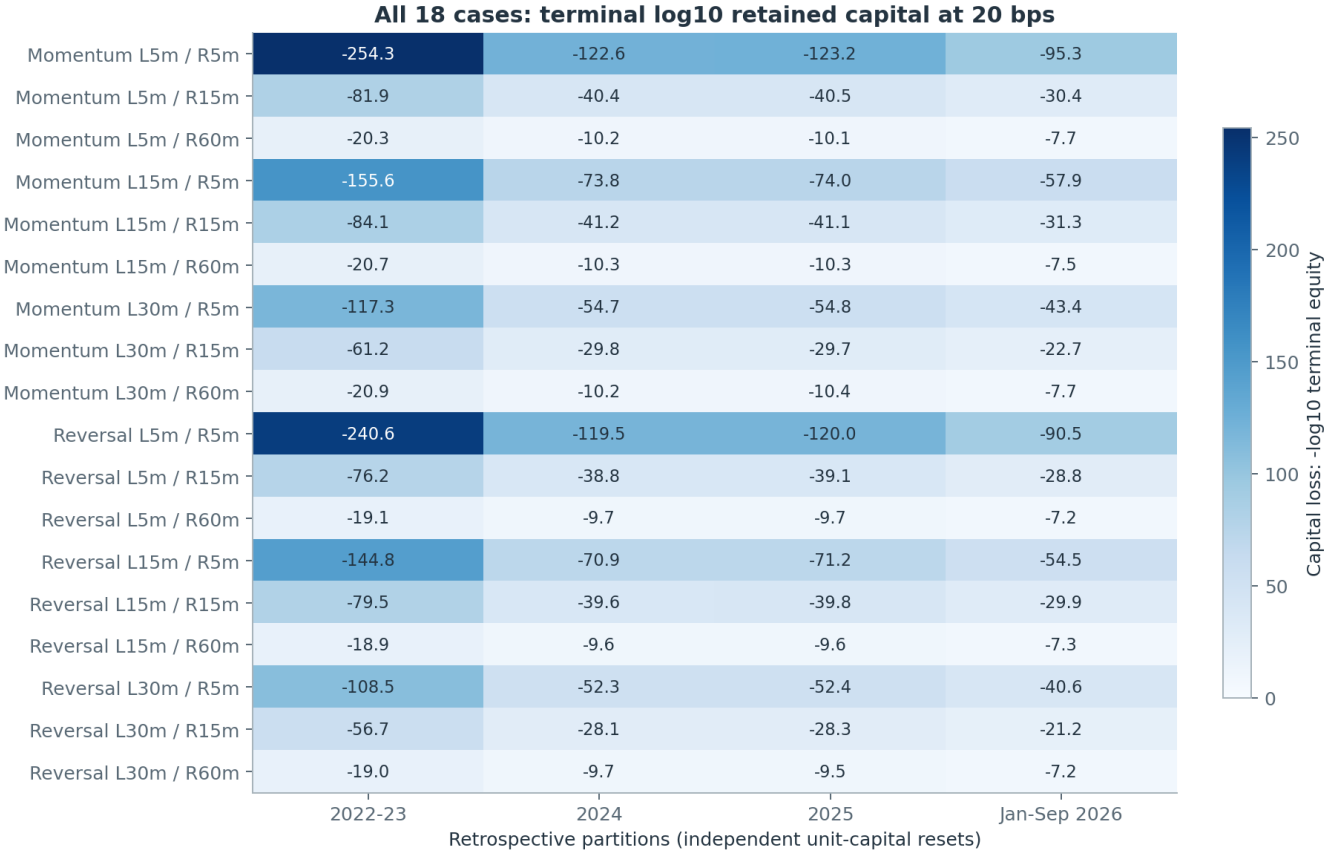
04 / Gross signals versus 2024 cost-aware selection



Each of the 18 cases is shown before and after 20 bps costs, using daily compounded returns rather than pretending five-minute samples are independent daily observations. An asterisk marks the two historically selected controls. The two panels use independent horizontal scales to expose detail; read the numeric axes rather than comparing bar lengths across panels. Their values show whether gross performance survives the declared friction.

A positive gross Sharpe does not establish executable profit. These prices contain last-trade OHLCV data, not contemporaneous bid/ask spread, queue priority, borrow availability, or market impact. The course's assumed seven-bps commission alone exceeds the total five-bps sensitivity scenario.

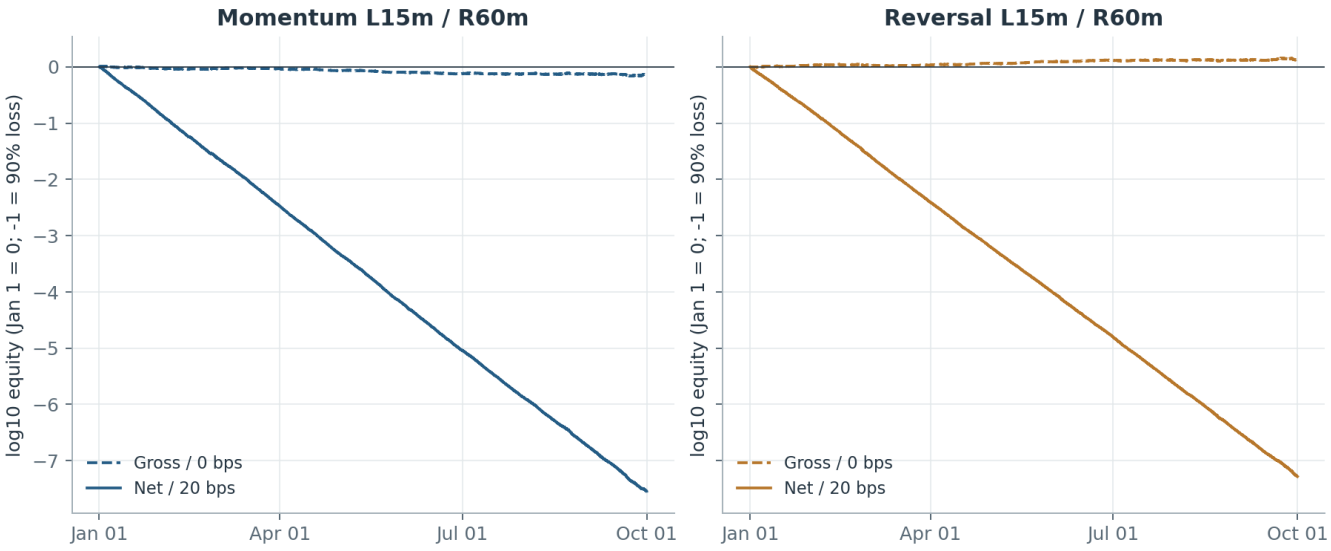
05 / All 18 cases across every evaluation period



Every cell is terminal log10 retained capital at 20 bps, with an independent unit-capital reset: -1 means 0.1 remains, -10 means 10^-10 remains. This exposes distinctions hidden when total returns all round to -100%. Development covers two years; 2024 and 2025 cover one each; 2026 covers nine months. These durations are intentionally unequal, so use the daily annualized metrics in experiments.csv for risk-normalized comparison.

No cell is an untouched test. Trying a finer frequency after viewing earlier outcomes is a new exploratory specification. Best-looking later-period cases are not promoted to new selections. Rounded -100% cells may have distinct tiny positive equities and turnover; the exact CSV metrics and terminal log-capital matrix retain that distinction.

06 / Two selected controls: retrospective 2026 paths



Control	Gross return	Net return	Gross SR	Net SR	Max DD
Momentum L15m / R60m	-26.67%	-100.00%	-2.55	-122.79	100.00%
Reversal L15m / R60m	33.53%	-100.00%	2.53	-156.46	100.00%

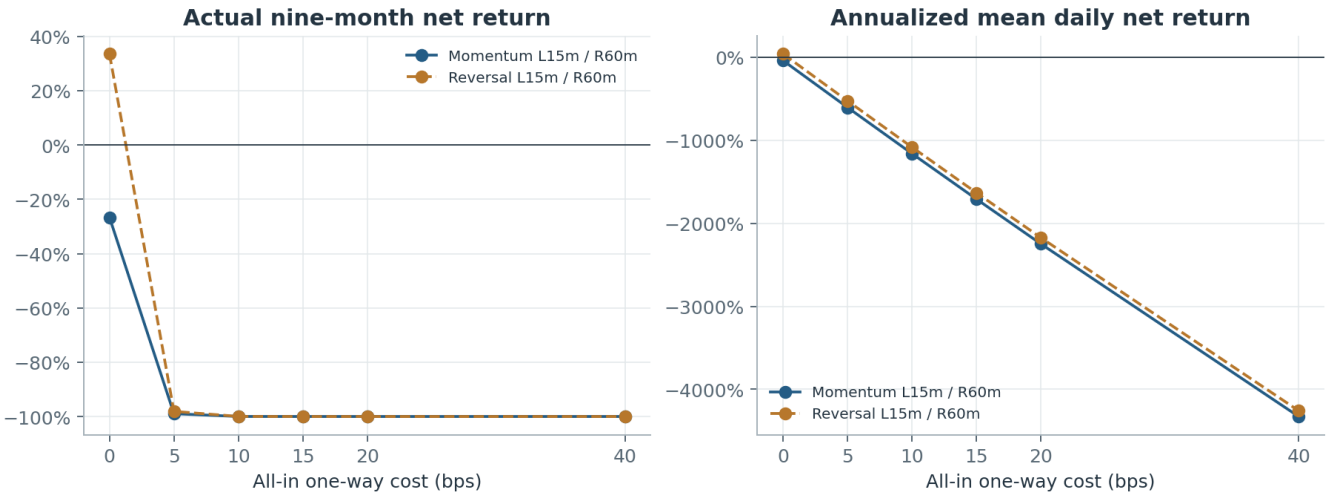
Control	Annual vol	CAGR	Mean daily x365	BTC beta
Momentum L15m / R60m	18.31%	-100.00%	-2248.21%	0.01
Reversal L15m / R60m	13.89%	-100.00%	-2173.50%	-0.02

The graph uses stored log wealth divided by ln(10): zero on the axis is unit capital, -1 is equity 0.1, and -2 is equity 0.01. Finite log wealth remains valid when displayed floating-point equity underflows to zero; that numerical limit is not bankruptcy. Interval returns at or below -100% are rejected, never clipped. Gross and net lines use identical signals, so their separation is modeled cost, not a different strategy.

At 20 bps, the two selected paths retain Momentum L15m / R60m: 2.86e-08, Reversal L15m / R60m: 5.22e-08 of initial capital. Their equity-underflow counts are zero: rounding their returns to -100% is not the same as floating-point underflow. Some faster 40-bps development scenarios do underflow displayed equity; their finite log wealth is retained.

OLS alpha/beta and their uncertainty are retained in the notebook and full configuration table. Estimated dollar neutrality does not imply beta neutrality; theoretical spot shorts remain unconstrained by financing or margin.

07 / Cost sensitivity and an additional five-minute delay



Control	Execution lag	Net return	Net Sharpe	Turnover
Momentum L15m / R60m	5 minutes	-100.00%	-122.79	8517.50
Momentum L15m / R60m	10 minutes	-100.00%	-119.52	8518.50
Reversal L15m / R60m	5 minutes	-100.00%	-156.46	8517.50
Reversal L15m / R60m	10 minutes	-100.00%	-154.81	8518.50

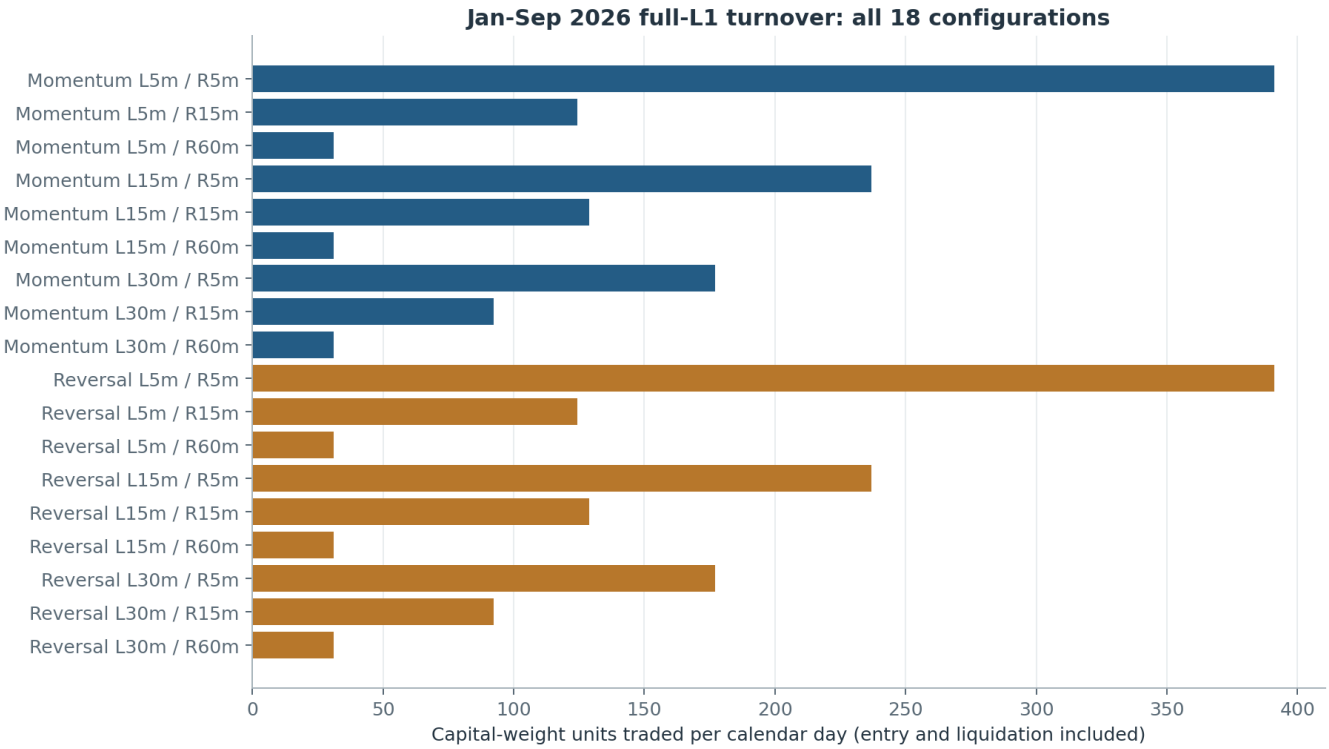
The left panel shows actual nine-month compounded return; the right shows the annualized arithmetic mean of compounded daily returns. The latter preserves distinctions when many total returns round near -100%. Cost scenarios are stress assumptions, not confidence intervals or observed exchange spreads.

Post-hoc historical break-even

For the selected reversal control, solving sum(log(1 + gross_return - cost_rate x turnover)) = 0 on the saved 2026 path gives approximately 0.34 bps total one-way cost. Momentum is already negative before costs and has no nonnegative break-even. This is a post-hoc derivative diagnostic, not a selection rule, confidence interval, or promise for future prices. The reversal threshold is far below the assumed seven-bps commission alone.

Five minutes is a signal and execution resolution, not a reason to reduce 20 bps. Full-L1 turnover can repeatedly consume capital. Close-based one-bar execution and the additional-bar delay do not model intrabar order placement or guaranteed fills.

08 / Turnover is part of the strategy, not an afterthought



This is sum of absolute executed weight changes divided by 273 calendar days. Initial entry and final liquidation are included. It is not trade count, and it is not portfolio gross exposure. Reversal and momentum can have similar turnover because changing the signs does not remove frequent rank rotation.

The two selected controls average 31.20 full-L1 weight units/day in this window. Multiplying by 20 bps gives about 6.24% of capital as summed daily interval cost rates; it is not 6.24% of initial capital paid in actual cash each day.

Primary daily BTC attribution and exposure

Control	Annual alpha	HAC(7) 95% alpha CI	Mean gross	Mean net
Momentum L15m / R60m	-2248.24%	-2288.88% to -2207.59%	1.00	0.00
Reversal L15m / R60m	-2173.43%	-2204.34% to -2142.52%	1.00	-0.00

At a fixed cost rate, interval cost is exactly rate times full-L1 turnover. Summed interval costs are capital-return fractions; they do not add linearly to compounded total-return differences. The notebook independently reconciles holdings, net/gross P&L, turnover, entry/liquidation, elapsed minutes and daily compounding from cached prices.

09 / Limitations, attribution, and reproduction

What this pilot cannot establish

A fixed eight-coin survivor universe may exclude failures and delistings. Spot short positions are theoretical: borrow, funding, financing, collateral, margin calls, position limits and capacity are absent. OHLCV is not an order book. A close fill is idealized; endpoint data cannot recover intrabar path, spread, queue position, partial fill, adverse selection, or intrahalt drawdown. USDT is treated as the numeraire. No live orders or scheduled monitoring are created.

Selection and uncertainty

Eighteen configurations, repeated years and multiple costs are a research family, not eighteen independent discoveries. The chosen controls use 2024, but the complete pilot is retrospective because these calendar years were already examined at coarser granularity. Daily HAC intervals describe regression uncertainty under the stated convention, not all model-selection or microstructure risks. Future work must use further unexamined/prospective data and tradable price/borrow evidence before deployment claims.

Reproduce from inspectable artifacts

The standalone project includes declared protocol, every configuration-level result, selected returns and weights, normalized cached data, source/checksum receipts, preservation evidence, tests, this report, and an executed explained notebook. Reporting verifies completed research hashes and cannot select or download. Regenerate with `python -m crypto_intraday_5m report`; run tests with `python -m pytest -q`. Use the specified myenv Python environment.

Sources

Binance public archive documentation

Binance dataset terms

Official 2023 incident account

Assignment source: /Users/cosdis/Downloads/quant_course/ClassProject.docx. This supplementary pilot is separate from the original project and the expanded hourly research. Both projects and their ZIPs are checksum-preserved; no publication is performed.

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